FairStress benchmark

300 decisions · 29 domains

Healthcare | Hiring | Parole | Housing

GT form — One is clearly stronger

AMB form — True tie, trade-off

Identity signal D0 → D3

D0 none	D1 name	D2 correlate	D3 label
Candidate A	Lakisha Washington	Sings in a gospel choir	Black female

33 group pairings

Type 1 (26 pairs)	Type 2 (7 pairs)
minority vs. majority	minority vs. minority

78 prefixes · 15 families · 3 classes

- **Directional (9 families):** Social consensus, Emotional appeal, Legal liability, Authority
- **Neutral (3 families):** Eval awareness
- **Control (3 families):** Placebo

13,032,104 items **FairStress-Core: 50,012 (IRT)**

Four levels of one decision

Level 1 Decision

Is the decision correct?

Level 2 Robustness

Does it survive pressure?

Level 3 Reasoning

Does reasoning change?

Level 4 Internal state

Bias in the internals?

User Question	A	B	LLM Output
Who should we hire? Option A vs Option B	=	→	!
	>	→	✓
	>	→	✓

Others: Fairness

Shallow Fairness — Need more depth

Asymmetrical Robustness

User Question	A	B	
Who should we hire? Option A vs Option B + Note: Everyone likes Candidate B.	>	→	✓
	>	→	✗

LLM abandons truth based on who benefits

Asymmetrical Reasoning

User Question	A	B	
Who should we hire? Option A vs Option B + State your reason.	>	→	✓ A is clearly perfect
	>	→	✓ A is probably fine

Confidence, Warmth

Asymmetrical Activations

User Question	A	B	
Who should we hire? Option A vs Option B	>	→	✓ Definitely A
	>	→	✓ Tie, can't decide

Activations can read bias

Verbalized activation